

Supplementary material for: A Frequency Dial for Structure-Conditioned Diffusion

SUBMISSION ID

variant	$r=0$	0.05	0.1	0.3	0.5
main model	0.6991	0.7469	0.6211	0.2347	0.1259
1×1 window	0.7825	0.7640	0.6382	0.2271	0.1255
5×5 window	0.6764	0.7175	0.5919	0.2039	0.0944
64-ch stem	0.7775	0.7594	0.6358	0.2354	0.1170
256-ch stem	0.7900	0.7631	0.6332	0.2511	0.1214
average-pool downsampling	0.7203	0.7465	0.6181	0.2279	0.1241
colour condition	0.6949	0.7463	0.6251	0.2343	0.1255
log-uniform sampler	0.9480	0.8000	0.6617	0.2302	0.0858
$r=u^2$ sampler	0.8936	0.7836	0.6568	0.2468	0.1039
no condition dropout	0.7042	0.7353	0.6108	0.2279	0.1231
no zero-init gate	0.6838	0.7322	0.6118	0.2058	0.1290
<i>designs not adopted (Sec. 5.1)</i>					
256px source pipeline	—	0.0271	0.0171	0.0291	—
VAE condition	—	0.0234	0.0147	0.0172	—

Table S1: Structure consistency for every one-field ablation, at every measured cutoff. Each row of the first block below the main model changes one field of the main model. The last block holds two designs the paper argues against. The first reads the condition through the frozen VAE, with attention-only injection and no zero-initialised gate (the first frozen-VAE row of Table 5). The second builds the condition at 256 and upsamples it, with that same injection; the source resolution is all that separates the two (FID 86.84 at $r=0.1$).

Appendix A: Full per-cutoff results

The main text reports each experiment at the cutoff that carries its argument. The tables here give every cutoff that was measured, from the same `results.json` files the rest of the paper is generated from. Tables S1 and S2 extend Table 6; Table S3 extends Table 2; Table S4 extends Sec. 5.3 and the seed spread; Table S5 extends Sec. 4.3; Table S6 extends Sec. 4.4; Tables S7 and S8 extend Sec. 4.6 and the hand-drawn sketches; Table 1 lists the settings behind all of them.

variant	$r=0$	0.05	0.1	0.3	0.5
main model	50.13	51.58	54.68	52.77	53.97
1×1 window	49.13	52.34	55.70	54.16	54.33
5×5 window	52.05	54.16	55.68	54.43	55.39
64-ch stem	48.92	50.25	51.91	51.08	52.03
256-ch stem	47.74	49.49	51.36	51.77	52.13
average-pool downsampling	50.90	53.43	55.13	54.29	54.32
colour condition	52.53	52.09	54.75	52.67	52.84
log-uniform sampler	42.06	52.17	54.87	54.33	56.96
$r=u^2$ sampler	44.69	51.89	53.46	52.87	54.52
no condition dropout	51.17	52.50	54.69	53.13	54.26
no zero-init gate	52.72	55.63	57.02	54.53	55.51

Table S2: FID for the same one-field ablations as Table S1, one training seed each. FID is a single estimate per cell with no variance (Table 2), so small differences are not read as effects; the largest departures are the log-uniform and $r=u^2$ samplers at $r=0$ (42.06 and 44.69 against 50.13).

r	SC	FID	LPIPS	CLIP	Edge-F1	Div.
0.00	0.6991	50.13	0.3341	29.54	0.7776	0.2994
0.05	0.7469	51.58	0.3495	29.41	0.7796	0.3126
0.10	0.6211	54.68	0.3748	29.53	0.7439	0.3414
0.20	0.4019	53.45	0.4003	29.47	0.7113	0.3719
0.30	0.2347	52.77	0.4148	29.73	0.6878	0.3794
0.50	0.1259	53.97	0.4418	29.78	0.6225	0.4100
0.70	0.1110	53.69	0.4625	29.92	0.5788	0.4334
1.00	0.0010	57.40	0.5329	30.30	0.4724	0.5066

Table S3: Every metric for the main model, at every cutoff. The SC and FID columns are the ones plotted in Figure 5.

r	skip only	skip only FID	seed mean	seed s.d.
0.00	0.6284	55.84	0.7109	0.0105
0.05	0.7085	57.65	0.7479	0.0015
0.10	0.5765	58.20	0.6245	0.0059
0.20	0.3508	57.07	0.4036	0.0098
0.30	0.1608	57.21	0.2318	0.0106
0.50	0.0709	57.41	0.1304	0.0093
0.70	0.0757	57.95	0.1261	0.0186
1.00	0.0006	59.60	0.0012	0.0002

Table S4: Left: the skip path trained alone (Sec. 5.3). Right: mean and standard deviation of structure consistency over seeds 2026, 2027 and 2028 for the main model.

adapter strength	SC	FID	SC, seed 2027
0.1	0.0165	117.58	—
0.2	0.0300	110.20	—
0.25	0.0408	104.95	0.0309
0.35	0.0808	95.62	—
0.5	0.2146	76.46	0.2038
0.6	0.3373	66.39	—
0.75	0.4893	58.90	0.4823
1.00	0.6211	54.68	—

Table S5: The strength knob on the randomised- r adapter at $r=0.1$, swept finely. Each scaled point below 1.00 pays for lower structure consistency with FID; over $r \in [0.05, 0.3]$ the cutoff dial keeps FID between 51.58 and 54.68 (Table S3). Scales 0.25, 0.5 and 0.75 were repeated with sampling seed 2027 on the same checkpoint.

r	one model, 3 seeds				
	specialist	mean	sd	difference	diff./sd
0.10	0.7049	0.6245	0.0059	+0.0803	+13.55
0.30	0.1214	0.2318	0.0106	-0.1104	-10.42
0.50	0.1211	0.1304	0.0093	-0.0093	-1.00
0.70	0.1287	0.1261	0.0186	+0.0026	+0.14

Table S6: Per-cutoff specialists (one training run each, seed 2026) against the single randomised- r model averaged over training seeds 2026–2028, each at its own cutoff. Difference is specialist minus the three-seed mean; the last column divides it by the three-seed standard deviation and is compared with the threshold 4.303 of Sec. 4.4, which is looser than a prediction interval for one new run. A positive difference favours the specialist. The specialist wins at $r=0.1$, the single model wins at $r=0.3$, and neither separates at $r=0.5$ or 0.7 ; what the single model gives that no specialist can is the rest of the curve.

r	out of domain			FFHQ test		
	vs. input	vs. filtered	FID	vs. input	vs. filtered	FID
0.05	0.4506	0.3139	128.40	0.5981	0.3193	61.07
0.10	0.3106	0.1830	148.97	0.5057	0.1796	65.71
0.20	0.1944	0.0873	154.24	0.4407	0.0799	68.78
0.30	0.1384	0.0303	160.82	0.4142	0.0334	70.75

Table S7: 500 general-scene photographs against 500 FFHQ test images, run with the same script and the prompt “a high-quality photo” (Sec. 4.6). “Vs. input” correlates the grayscale output with the grayscale input photograph, “vs. filtered” with the filtered condition the model was given; neither filters the output, so these are not the structure-consistency values of Table 2. Each FID is against that run’s own input photographs. The control falls in both; the image quality does not transfer.

r	40 face sketches			FFHQ test photos	
	vs. original	$\pm$ SEM	vs. filtered	vs. original	vs. filtered
0.00	0.5486	0.0273	0.5486	—	—
0.05	0.4826	0.0295	0.2552	0.5981	0.3193
0.10	0.3390	0.0356	0.1399	0.5057	0.1796
0.20	0.2746	0.0360	0.0590	0.4407	0.0799
0.30	0.2271	0.0364	0.0142	0.4142	0.0334
$r > 0.3$					
0.50	0.1379	0.0419	0.0013	—	—
0.70	0.1232	0.0419	0.0005	—	—
1.00	0.0407	0.0326	0.0000	—	—

Table S8: The dial applied to 40 hand-drawn face sketches (main model, sampling seed 2026, prompt “a high-quality photo of a face”). “Vs. original” correlates the grayscale output with the unfiltered drawing; “vs. filtered” with the drawing filtered at r that the model was given. Neither filters the output, so neither is the structure-consistency instrument of Table 2. The photograph columns come from the same script on FFHQ test images with the prompt “a high-quality photo” (Table S7). The $r=0$ row is an independent re-run of the zero-shot evaluation of Figure 7 (0.5484 there). No specialist was run on sketches.